

The Silent Force Multiplier: The History and Role of UAVs in Warfare

Lt. Kendra L. B. Cook
United States Air Force
Wright-Patterson AFB, OH
937-257-6600
Kendra.Cook@wpafb.af.mil

Abstract—Many don't realize that the history of UAVs started nearly a century ago, and that the modern era of UAVs goes back nearly four decades. It often goes unnoticed that UAV development and employment is being pursued by more than 50 countries world-wide.

This paper examines the history of unmanned aerial vehicles including the initial concepts and employment and the role of UAVs in warfare, providing examples from several conflicts throughout the world. An analysis of the reasons for the underutilization of UAV capabilities, the reasons why UAV missions have been successful, and basis for continuing to fund UAV development and employment is given. Finally, current UAV trends are discussed.^{1,2,3}

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1. INTRODUCTION

Unmanned aerial vehicles (UAVs), such as the United States' RQ-1 Predator and RQ-4 Global Hawk, have flown numerous successful missions since the start of Operation Enduring Freedom (OEF) and Operation Iraqi Freedom (OIF). To many Americans, the idea of using unmanned aerial vehicles as a tactical weapon or a military force multiplier is a new concept. Actually, the history of UAVs started shortly after the first manned flight, and the modern era of UAVs goes back nearly 4 decades. The lack of public visibility about the accomplishments of UAVs, particularly during the past 40 years, and the misunderstanding of the applications of these systems are key reasons why UAVs were not exploited by military planners until the mid-1980s. For example, the UAV missions flown in SE Asia during the Vietnam Conflict were highly classified and few people, including the military members themselves, were aware of

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the role these systems played.

UAV technology is being pursued by more than 50 countries world-wide, however, the use of UAVs by foreign nations is not widely publicized with few exceptions such as Hizballah's recent use of the Ababil II UAV against Israel and the use of surveillance and reconnaissance UAVs by coalition partners in OEF and OIF.

This paper examines the history of unmanned aerial vehicles including the initial concepts and employment, the role of UAVs in warfare including: WWII, the Cuban Missile Crisis, the Vietnam Conflict, the 1982 Israeli-Lebanon Conflict, Operation Desert Storm and Operation Desert Shield, Operation Enduring Freedom and Operation Iraqi Freedom, and the 2006 Israeli-Lebanon Conflict; as well as the current and future threat of the use of UAVs by terrorist groups. There are many other instances of UAVs taking an active role in warfare by other countries, however this paper focuses primarily on American systems.

As will be shown, historical missions of UAVs include: Intelligence, Surveillance and Reconnaissance (ISR); Force Protection; Suppression of Enemy Air Defense (SEAD); Counter Narcotics, Psychological Operations, All Weather / Night Strikes, Exercise Support and Anti-Submarine Warfare [14].

An analysis of the reasons for the underutilization of UAV capabilities, the reasons why UAV missions continue to be successful, and basis for continuing to fund UAV development and employment is given. Finally, current UAV trends are discussed.

This paper will illustrate that in wartime UAVs have been heavily utilized; but in peacetime, when their specialized and high-risk missions are not conducted, funding has disappeared. By continuing to fund and develop UAVs in peacetime, and by properly integrating them into the military, UAVs will continue to serve as an effective force multiplier and allow lives to be saved in future conflicts.

2. THE EARLY YEARS: 1916 - 1958

The history of UAVs started shortly after the first manned flight – the origins of UAV development can be traced back to 1916 when Elmer Sperry of the Sperry Gyroscope

Company (along with Peter Hewitt) successfully developed an automatic control system for the Curtiss Flying boat. After U.S. entry into WWI in 1917, funds were made available for the development of a flying bomb. The US Navy ordered five aircraft designed as flying bombs. During the Navy trials, which began in late 1917, the unmanned Curtiss N-9 seaplanes suffered crashes after catapult launch and engine failures. Despite extensive aircraft redesign, the program suffered more setbacks resulting in program cancellation in 1918.



Figure 1. Henry Ford of the Ford Motor Company and Glenn Curtiss in front of the Curtiss Flying Boat 1913 [2]

Despite setbacks in the Navy's flying bomb program, the U.S. Army was interested in continued flying bomb development and gave a contract to Charles Kettering to develop an unmanned aircraft. Twenty-five of the small, cheap, crudely built biplanes named the "Kettering Bug" were ordered in 1918. Successful flights were entwined with crashes, but the successes were dramatic enough for the Army to purchase an additional 75 aircraft. The longest flight made by a "Kettering Bug" was only 16 miles. It was never flown operationally during its lifetime [2].

Meanwhile, in mid-1917 in the United Kingdom, Farnborough built six pilot-less aircraft and tested three of them at the Northolt Aerodrome near London. The Northolt experiments involved launch of an aircraft from a 50-ft horizontal ramp and up to 100-ft incline in free flight, but each aircraft stalled and crashed. Around the same time, Germans were also experimenting with unmanned gliders that weighed up to 2,205-lbs and had a range of five miles after being launched from airships at Jutersborg [9].

1917-1945 World War II

During WWII, the Germans developed a simple unmanned aircraft known as the V-1 "Buzzbomb". Widely heralded as the first successful cruise missile, thousands of these unmanned aircraft performed one-way missions flying at

nearly 400 mph and less than 1,000 feet off of the ground.



Figure 2. German V-1 "Buzzbomb" in flight [12]

At this speed, the flight from France to London took a mere 22 minutes and had devastating effects. 8892 bombs landed in the UK causing more than 6,200 casualties and seriously wounding 18,000 others. The damage caused by these inexpensive and relatively simple weapons had been traumatic.

3. THE MODERN ERA: 1959 - PRESENT

The modern UAV era began in late 1959. Prompted by the shoot-down of Gary Francis Power's U-2 over Russia, the first modern UAV efforts consisted of placing cameras in target drones. It was meagerly funded, but the hope was to minimize the risk to pilots and crew, as well as to expensive airframes, from flying aircraft over hostile territory.

In July 1960, the U.S. Air Force awarded a \$200K contract to Ryan Aeronautical Company for a project, code named "Red Wagon", to demonstrate a modified "Firebee" target drone that would perform remote-controlled photographic surveillance missions [9]. The radar cross section was reduced on the "Red Wagon" by placing a wire screen over the jet intake, painting parts of the drone with non-conductive paint, and placing radar-absorbing pads on both sides of the fuselage. These modifications did not compromise the Firebee's performance. In 1961, when the Kennedy Administration took office, many military projects, including the Red Wagon, were put on hold.



Figure 3. BQM-34A "Firebee" drone

1962 Cuban Missile Crisis

Following the October 1962, Cuban SA-2 Surface-to-Air Missile (SAM) downing of a manned United States Air Force U-2 over Cuba, Department of Defense officials frantically inquired as to the availability of the Red Wagon Drones. However, due to program cancellations by the Secretary of Defense for Research and Engineering, only two drones had been built. In the wake of their unavailability during the Cuban Missile Crisis, U.S. UAV programs would finally receive the necessary momentum to ensure UAVs were ready for action.

1964-1972 Vietnam

The first substantial use of UAVs in the United States was during the Vietnam War. From 1964 to 1972, over 5,000 Americans lost their lives in SE Asia in downed aircraft and nearly 90% of the American POWs were pilots or crewmen who had been captured. Unbeknownst to most, during the same time period, over 3,400 combat UAV sorties were flown by SAC's 100th Strategic Reconnaissance Wing over North Vietnam, China, Laos, and other locations in Southeast Asia. These UAVs returned from missions deep within enemy territory at a fraction of the cost of manned reconnaissance aircraft, and without the threat to American personnel.

The principal UAVs flown during the war in South East Asia were the "Lightning Bug" and later the "Buffalo Hunter", both made by Ryan Aeronautical Company. These UAVs suffered an attrition rate of less than 10% during the Vietnam Conflict while performing photographic reconnaissance missions, battle damage assessments (BDA), conducting electronic intelligence (ELINT), and distributing propaganda leaflets [5].

1982 Israel / Lebanon Conflict

In 1981, a Government Accountability Project (GAP) report accused the U.S. military of not sharing the high regard for UAVs that the Israelis had. The committee report professed it "would also like to convey support for the requirement to have UAVs in our military in view of their demonstrated success in actual combat". It also stated, "Increased use of UAVs in high risk and political situations could greatly reduce losses in future conflicts."

Between 1978 and 1982, cross-boarder conflicts between Israel and various forces in Southern Lebanon occurred at various levels of intensity. In 1982, after building tension, a landmine killed an Israeli officer and the rocket attacks and air strikes by the Israelis against Lebanon recommenced [4].

Historically, the Lebanese had used SA-6 surface-to-air missiles to destroy Israeli aircraft. In response, during the 1982 conflict, the Israelis used unmanned vehicles to harass the Lebanese, saturate their air defense system, deplete their

missile supply, and screen Israel's fighter aircraft. The SA-6s were decoyed by the US-designed Samson air launched UAVs which simulated incoming strike aircraft. Overall, this strategy proved extremely effective for the Israelis.

1990-1991 Operation Desert Storm and Operation Desert Shield

The Israeli-made Pioneer UAV flew over 300 combat reconnaissance missions with the U.S. military during Persian Gulf operations in 1990-91. The RQ-2A Pioneer system received extensive acclaim for outstanding performance by Army, Navy, and Marine Corps commanders for its effectiveness as a Reconnaissance /Surveillance/Target Acquisition (RSTA), Naval Gunfire Support, BDA, and battlefield management platform.



Figure 4. Pioneer UAV [6]

During the Gulf War all the units operating UAVs had individuals or groups attempt to signal the Pioneer at various times, possibly to indicate a willingness to surrender. One notable incident occurred when the USS MISSOURI, using a Pioneer to spot 16 inch rifle fire, overwhelmed the defenses of Faylaka Island off the coast near Kuwait City. Shortly thereafter, while still over the horizon and invisible to the defenders, the USS WISCONSIN sent another Pioneer UAV over Faylaka Island at low altitude. When the UAV flew over the island, it was audible to the Iraqis, as the air vehicle was intentionally being flown low to let them know that they were being targeted. The Iraqis recognized that the UAV overhead meant there would soon be more 2000-lb naval gunfire rounds landing on their positions with the same accuracy. The Iraqis made the right choice and, using handkerchiefs, undershirts, and bed sheets, they signaled their desire to surrender [5, 11].



Figure 5. Pioneer UAV aboard USS Missouri [6]
2001 – 2006 Operation Enduring Freedom and Operation Iraqi Freedom

The RQ-4 Global Hawk, manufactured by Teledyne Ryan, a subsidiary of Northrop Grumman, was designed to give the USAF an overwhelming lead in intelligence-gathering in the next century, and has proved its worth in both OEF and OIF. This platform has a 116-ft wingspan, and its airframe is largely made of composite materials. It was originally designed to fly autonomously from a base in the U.S. to a country of interest, loiter for 24-hours at a speed of 400-mph and an altitude of 65,000-ft while transmitting surveillance data, and then return to its base without refueling [13]. The RQ-4 Global Hawk is not intended for use over high-threat areas, but with its payload consisting of state-of-the-art EO/IR and SAR sensors, it can stand off a considerable distance while still providing both wide-area and spot imagery from very high altitudes.



Figure 6. Global Hawk

The Global Hawk UAV was deployed to the Afghanistan theatre and performed reconnaissance missions in support of Operation Enduring Freedom (OEF). Global Hawk provides Air Force and joint war-fighting commanders near-real-time, high-resolution intelligence, surveillance and

reconnaissance (ISR) images, while flying more than 60 combat missions and logging more than 1,200 combat hours to date[3].

Global Hawk UAVs have also flown missions in Operation Iraqi Freedom (OIF), providing numerous images to U.S. intelligence agencies as well as to all levels of each component of the military. Although they represented only a small percent of all the airborne image collection missions flown during the war, Global Hawks generated over half of all time-critical data on air defense targets. One particular Global Hawk in theater located at least 13 surface-to-air missile batteries, 50 SAM launchers, 300 canisters and 70 missile transporters; it also imaged 300 tanks, 38 percent of Iraq's armored force - a remarkable display of the vehicle's capability. The Joint Forces Air Component Commander (JFACC) credited Global Hawk with accelerating the defeat of the Iraqi Republican Guard, shortening the duration of the war and reducing casualties, exceeding the combatant commander's expectations [3].

During OIF, the Air Force also developed a full "reach back" capability for the Global Hawk, in which the UAV and its sensors were operated remotely from Beale Air Force Base, California, reducing Global Hawk's logistical footprint in the field by more than 50 percent. Global Hawk crews used Internet-style chat rooms to stay in touch, literally forming "a worldwide virtual crew." These chat rooms provided effective command and control over a weapon system that was spread across the globe [3].

The RQ-1 Predator UAV is manufactured by General Atomics and first flew in 1994⁴. It was designed to be a joint-service UAV for the Pentagon. The Predator has an advertised endurance of 40 hours and cruises at an altitude of 25,000-ft. Its principle payload is the electro-optic sensor turret, incorporating a mix of daylight, infrared and laser sensors. The RQ-1 sensor payload may also include a synthetic aperture radar (SAR), providing a means to survey up to 1,300 nautical square miles.

Hellfire-missile-packing RQ-1 Predators have also flown combat missions in Afghanistan and Iraq. "People just started opening their eyes to the capability of the aircraft," Capt. Sam J. Vanzanten (Predator controller) explained. He also noted that the Predator can conduct multifaceted roles over the battlefield. Even if the Predator isn't armed, "we at least have a camera and a radio to 'talk' to other aircraft in the area," he noted, to communicate target locations to pilots of conventional warplanes [7].

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⁴ The Predator flew its first operational mission over Bosnia in 1995.



Figure 7. Predator UAV [3]

When it's not firing Hellfire anti-tank missiles at the enemy, the Predator uses its powerful surveillance cameras to give the theater air component commander continuous real-time surveillance of the battlefield.

2006 Israeli-Lebanon Conflict

In August 2006, the Lebanese Hizballah launched three Iranian-made Ababil II UAVs in attempts to infiltrate Israeli airspace. The Israeli Air Force shot down one of the Hizballah UAVs in a night-time air-to-air engagement off the coast of Akka in northern Israel on 7 August 2006. The external configuration of the UAV shot down is identical to that of the Iran Aircraft Manufacturing Industries (HESA) Ababil-T Lethal UAV, strengthening Hizballah claims made since 2004 that it is able to field an attack UAV. Israeli Defence Forces claim that the "...UAV was Iranian". The Ababil-T in its standard configuration carries a daylight television camera as well as a medium sized high explosive warhead. The UAV was flying at night, indicating that it was either flying using pre-programmed GPS waypoints or its sensor package has been modified to include a thermal imaging system [1].

The Hizballah-launched UAV was engaged approximately 10 km offshore by a Israeli Air Force F-16. The engagement "lasted less than ten minutes," according to the IDF, "including the identification and interception stages". The Israeli Air Force's "northern air control unit" reportedly identified the craft before it crossed the border and dispatched aircraft to intercept it. The UAV was shot down at a low altitude over the Mediterranean Sea [1].

Current and future threat of UAVs by terrorist groups

UAVs are a very attractive option for terrorists anxious to deliver a covert attack. Even armed with conventional explosives, UAV can become more efficient than suicide terrorists. Substantial damage can be inflicted, in particular, if a UAV is directed to a crowded place. In addition, UAVs can penetrate areas where suicide terrorists have almost no chance of reaching [8].

Experts point to a set of advantages that may make UAVs seem attractive to terrorists [10]:

- They are able to attack targets that are difficult to reach by land (targets limited to cars loaded with explosives

or suicide terrorists).

- There is the possibility of carrying out a wide-scale (area) attack, aimed at inflicting a maximum death rate on a population (particularly, through the use of chemical or biological weapons in cities).

- UAVs offer covert attack preparation and flexibility in choice of a UAV launch site.

- There is also the possibility of achieving a long range and acceptable accuracy with relatively inexpensive and increasingly available technology.

- There is a relative cost effectiveness of UAVs compared with ballistic missiles and manned airplanes.

- UAVs can achieve a strong psychological effect by terrorizing people and putting pressure on politicians.

UAVs are inexpensive and easily proliferated, and therefore are considered to be a potential weapon of choice for terrorist groups world-wide. Many terrorist groups have expressed interest in obtaining UAVs to use as weapons. Groups such as the Lebanese Hizballah have acquired and launched UAVs against their enemies. While these flights typically have not resulted in casualties, they still cause panic and fear: one of the results terrorists are looking for.

Examples of terrorists attempting to use UAVs include [15]:

- In 1995, Aum Shinrikyo, the Japanese terrorist group that attacked the Tokyo subway with sarin gas, planned to use remote-control helicopters to spray dangerous chemicals from the air. However, the helicopters crashed during testing and were not used in the attack.

- In 2001 Osama bin Laden considered using remote-control airplanes packed with explosives to kill President George W. Bush and other heads of state at the G-8 summit in Genoa, Italy.

- In June 2002, quoting a German intelligence official, the *Reuters* news agency reported that al Qaeda might be planning to attack passenger aircraft using model airplanes.

- The Revolutionary Armed Forces of Colombia, or FARC, were discovered in possession of nine remote-controlled unmanned aircraft when a Colombian army unit overran one their remote camps in August 2002.

- According to *Reuters*, in early March 2004, Israeli intelligence prevented a terrorist act with use of a UAV loaded with explosives. Representatives of the administration of Prime Minister Ariel Sharon claimed that a Palestinian extremist group planned to attack a Jewish settlement in Gaza sector.

4. ANALYSIS

Why UAV assets are underutilized

Although UAVs offer enormous potential, integrating them into militaries has been an uphill battle. There are several

reasons for this. Initially there were constant problems with designing and controlling UAVs. Miniaturization of accurate components for midcourse guidance, digital signal processors, and sensor resolution are among the many technical hurdles which have limited UAVs role in the military up until the last five years. In addition, at the time when UAVs were able to successfully complete flights, there was a lack of appreciation regarding the benefits that UAVs could provide. Decision-makers were not aware of the capabilities and operational potential of unmanned air vehicles.

Another contributing factor in the lack of UAV development in the military is that UAVs directly compete with manned aircraft for funds that are already stretched. UAVs, and the missions they perform have historically been thought of as supplemental to the capabilities of manned aircraft and space based platforms, instead of the asymmetric and unique target identification and strike capabilities that are being demonstrated today. Finally, UAV proponents will continue to be confronted with military decision-makers that are comfortable with the proven abilities and performance of manned aircraft. With many of the high-level military personnel being former pilots, the decision: “manned vs. unmanned” may be very difficult.

Why UAV missions have been successful

UAV missions have been successful for several reasons. The technology that is being incorporated into the UAV systems is continually advancing. State-of-the-art technologies such as Synthetic Aperture Radars, increasingly capable microprocessors, increased data-link rates, radar-absorbing materials, the use of high bandwidth communications, and SATCOM-equipped navigation systems, are being integrated onto the platforms making them a key asset to militaries world-wide. Another key reason for UAV mission success is the UAVs’ low flying altitude and slow speed that makes them difficult for traditional enemy sensors to detect or recognize. UAVs may not be limited to the operating restrictions placed on manned aircraft – they have been sent on missions over enemy territory, against sophisticated integrated air defense systems – missions that would have to be thought twice about for manned aircraft due to cost or liability. With UAV operations, loss of human life is not a consideration making the decision to perform a high-risk mission easier.

Why we should continue to fund UAV development

UAVs are cost effective when the threat is high. The “threat” could be the actual threat to the aircraft, pilot or crew. In general, UAVs cost only a fraction of a manned aircraft. In conflicts such as Vietnam, where 90% of POWs

were pilots or crewmembers, it is absolutely ideal to send unmanned aircraft in to perform missions such as surveillance and reconnaissance. The “threat” could also be the threat of sensitive political or economic situations. UAVs are ideally suited to perform missions that cannot be economically or feasibly satisfied with manned aircraft. UAVs provide a viable alternative to manned aircraft.

Innovative uses for UAVs continually emerge. For example, in addition to tactical and strategic military functions, UAVs are also being used to perform border and coastal patrol missions, and are employed in the War Against Drugs in numerous countries. These civilian roles are offsetting the cost for the military and government contractors who are developing them.

UAV Trends

Several trends can be seen in UAV operations. First, they are continually advancing as history has shown. This includes advancing payload capabilities and incorporating low- observable technology. The capabilities of the aircraft themselves are ever increasing. In the early years of UAV development, UAV flights lasted less than 20 minutes. Now, UAVs can fly tens of thousands of miles and stay airborne for several days. Emerging technologies such as onboard intelligence, “sense and avoid” capabilities, SAR enhancements, multi-spectral imagery, Light Detection and Ranging (LIDAR) imaging, nuclear detection systems, sensor autonomy, lightweight and efficient power supplies, and mass data storage, are all improving the possibilities for future UAVs.

Second, trends in autonomy can be seen as shown in Figure 8. Current UAVs fly entire missions with little or no human intervention. The ultimate goal is to be able replace a pilot with a machine of equal or superior thinking speed, memory capacity, and responses gained from training and experience. This trend is being driven by advancements in commercial microprocessor capabilities. Today’s supercomputers are within a factor of 10 of achieving human equivalence in speed and capacity and could achieve human parity by the 2015 timeframe [14].

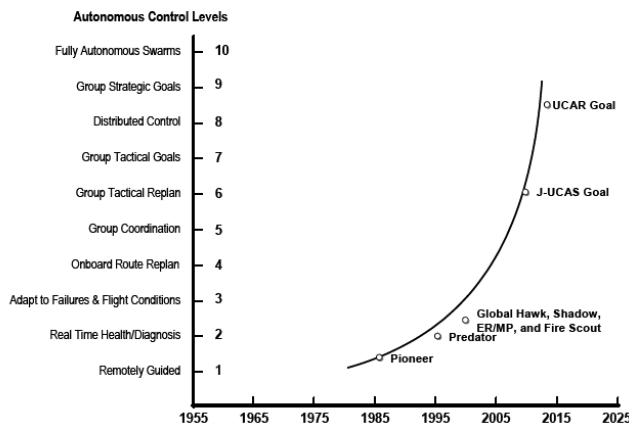


Figure 8. Trend in UAV Autonomy [14]

Third, they are becoming an integral part of military operations. Entire platoons and divisions are being dedicated to UAV operations in every branch of the military. Each service is beginning to recognize the value added when including UAVs in their operations. According to the 2005 DOD UAV Roadmap, The U.S. UAV inventory is expected to grow from 250 today to 675 by 2010 and 1400 by 2015 (not including micro and mini UAVs). The US Department of Defense is currently spending approximately \$2Billion on UAV RDT&E and procurement [14].

Next, unmanned combat aerial vehicles (UCAVs) are in development by a number of nations. These systems will be multi-role platforms capable of ISR missions, air-to-air and air-to-ground missions. While the U.S. operates the weaponized⁵ RQ-1 Predator UAV, the funding for a joint UCAV (JUCAF) was recently cut, effectively ending the program.

Finally, UAV technology is becoming cheaper and more widely proliferated, resulting in nations which previously could not afford UAV technology, gaining access to this effective force multiplier.

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⁵ Weaponized UAVs are those that have been modified after manufacturing in order to carry weapons as a payload. UCAVs are UAVs designed and built specifically to carry weapons.

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BIOGRAPHY

Kendra L. B. Cook is a lieutenant in the United States Air Force currently stationed at Wright-Patterson Air Force Base, OH. She received a B.S. in Aerospace Engineering from Boston University in 2004 and a M.S. in Astronautical Engineering from the Air Force Institute of Technology in 2006.